平滑化位相分散に基づいた敵対的音声プライバシー保護

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概要：生成 AI および音声合成技術の普及に伴い、音声プライバシーの保護が重要な課題となっている。従来の話者匿名化手法は主に振幅スペクトログラム変換やピッチシフトに依存しており、位相的特徴を抽出する最新の自動話者照合システムに対する頑健性が不十分である。本研究では、平滑化位相分散と非線形周波数ワーピングを統合した話者匿名化フレームワークを新たに提案する。提案手法は、位相の滑らかな変化に対する人間の聴覚的マスキング特性を利用し、位相スペクトルにガウス平滑化ノイズを注入することで、ASV が依存する時間的コヒーレンスに摂動を与える。同時に、非線形ワーピングを用いた振幅スペクトルの変調により、声道特性の変化を模擬する。LibriSpeech データセットおよび ECAPA-TDNN を用いた評価実験の結果、本手法は自動音声認識の単語誤り率の悪化を最小限に抑えて言語的有用性を維持しつつ、ASV の照合性能を著しく低下させ、堅牢な個人情報保護を実現した。

キーワード：音声プライバシー保護、自動話者照合、位相分散、敵対的攻撃

Adversarial Voice Anonymization using Smooth Phase Dispersion

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Abstract: With the widespread adoption of voice biometrics, protecting speaker privacy has become a critical challenge. Conventional voice anonymization techniques primarily rely on magnitude spectrogram transformations or pitch shifting, which remain vulnerable to state-of-the-art automatic speaker verification (ASV) systems that exploit fine-grained temporal and phase characteristics. To address this vulnerability, we propose a novel speaker anonymization framework that integrates smooth phase dispersion and non-linear frequency warping. Our approach leverages human auditory insensitivity to smooth phase alterations, deliberately injecting Gaussian-smoothed noise into the phase spectrum to disrupt the temporal coherence relied upon by ASV models. Concurrently, non-linear warping alters the magnitude spectrum to simulate distinct vocal tract modifications. Experimental evaluations using the LibriSpeech dataset and ECAPA-TDNN architecture demonstrate that the proposed method significantly degrades ASV performance, effectively preventing identity leakage, while maintaining high linguistic utility with minimal Word Error Rate degradation in automatic speech recognition.

Keywords: Voice Anonymization, Automatic Speaker Verification, Phase Dispersion, Adversarial Attack

1. Introduction

1.1 Background and Purpose Statement

The ubiquitous deployment of voice-driven interfaces and conversational artificial intelligence has seamlessly integrated speech technologies into daily life. However, this convenience comes with unprecedented privacy risks[1][2][3]. Speech signals carry rich, immutable biometric footprints that can uniquely identify individuals, exposing them to potential misuse, profiling, and surveillance[4][5]. With the rapid advancement of deep learning-based Automatic Speaker Verification (ASV) systems[6], eavesdroppers can effortlessly extract speaker identities even from short audio snippets. In light of stringent

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global data protection regulations, such as the GDPR, developing robust voice anonymization techniques has become an urgent imperative. The fundamental challenge in this domain lies in achieving an optimal trade-off: thoroughly concealing the speaker's identity for voice privacy while strictly preserving the linguistic utility of the speech for downstream tasks like Automatic Speech Recognition (ASR)[7].

The purpose of this research is to signify the establishment of an advanced voice alteration technique capable of intentionally deceiving machine-based biometric models while preserving linguistic naturalness for human listeners. The implications of establishing such acoustic and mathematical perturbation approaches extend far beyond mere user privacy protection. This paper directly contributes to elucidating the adversarial mechanisms—how malicious users employ deepfakes and sophisticated voice spoofing to bypass state-of-the-art detection and authentication systems. Clarifying the fundamental nature of these vulnerabilities provides an essential foundation for developing robust countermeasures against disinformation and enhancing the overall security of future voice-based systems.

1.2 Conventional Works: Pitch Shift and McAdams

In this paper, we focus on both the voice privacy of speech anonymization and the linguistic utility evaluated against an ASR system. Privacy protection performance is evaluated using the Equal Error Rate (EER), and Linguistic Utility is measured using the Word Error Rate (WER) via ASR [8].

Various signal processing techniques have been proposed to tackle this privacy-utility trade-off, with Pitch Shift and the McAdams coefficient method being the most prominent baselines[9]. Pitch shifting alters the fundamental frequency by linearly scaling the entire frequency spectrum. While its primary advantage is simplicity and the high preservation of linguistic utility with a low WER, it is highly vulnerable to modern neural ASV models, which can easily map the shifted frequencies back to the original speaker's latent space, resulting in inadequate privacy protection[10][11][12]. Conversely, the McAdams method modulates the spectral envelope to simulate a modified vocal tract length. Although this provides a stronger defense against ASV systems, its iterative resampling and shifting processes inevitably introduce severe acoustic artifacts. This inherent drawback leads to unnatural-sounding audio and a catastrophic degradation in ASR performance, rendering it impractical for real-world applications where speech intelligibility is paramount.

1.3 Main Concept and Novelty of Our Proposal

To overcome the fundamental limitations of these conventional approaches, we propose a novel hybrid voice anonymization framework. Rather than heavily distorting a single acoustic property, our approach strategically distributes minimal adversarial perturbations across the multiple feature dimensions that ASV models rely upon. Specifically, we integrate micro pitch shifting, non-linear frequency warping, and smoothed phase dispersion to simultaneously obfuscate the fundamental frequency, magnitude (formants), and temporal fine structure.

1.3.1 Preservation of Utility through Physical Acoustic Consistency

The fatal degradation of WER in the McAdams method is likely attributable to its independent and contradictory manipulation of pitch (based on the fundamental frequency) and vocal tract characteristics (represented by formants). This approach generates physically unrealistic acoustic profiles defying natural laws (e.g., possessing a "high fundamental frequency combined with an extremely long vocal tract"), which severely disrupts the inference processes within the acoustic model of the ASR system, e.g., Whisper [13][14].

In this paper, the primary factor enabling the proposed method to suppress the WER deterioration is the "synchronized downward shift" algorithm. In this approach, a non-linear frequency warping ($\alpha = 1.02$) is synchronized in the same downward direction as the downward shift of the fundamental frequency by -4.0 semitones. By mathematically mimicking the physical laws of the human vocal apparatus (specifically, the strong positive correlation between vocal fold size and vocal tract length) and maintaining acoustic physical consistency, the ASR system processes the input not as unknown, unnatural machine noise, but as a plausible voice of a different real speaker. Consequently, the destruction of linguistic information is minimized.

1.3.2 Band-Limited Phase Perturbation

Furthermore, the band limitation of the phase perturbation, introduced as a core component of the hybrid anonymization, significantly contributes to the mitigation of WER degradation. Generally, phase randomization destroys the continuity of the speech waveform, drastically

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reducing the system's ability to distinguish consonants during speech recognition. In this study, phase masking ($\beta = 0.01$) was applied exclusively to the low-frequency band below 2000 Hz, which the ASV model heavily relies upon for extracting speaker characteristics. This mathematically eliminates phase interference in the high-frequency bands that are essential for consonant recognition. Through the aforementioned acoustic and mathematical techniques, the proposed method prevents the excessive loss of linguistic information and achieves an optimal Pareto improvement in balancing privacy protection and speech utility.

Our main contributions are summarized as follows:

- **Design of a Comprehensive Hybrid Attack:** We introduce a novel hybrid adversarial framework designed to efficiently perturb all three independent acoustic features: F0, formant trajectories, and phase coherence. This multi-dimensional disruption ensures robust obfuscation against state-of-the-art ASV systems.
- **Overcoming the Privacy-Utility Trade-off:** By restricting the magnitude of modifications and relying on smoothed phase noise, our method achieves a high EER comparable to aggressive conventional methods, while maintaining a significantly low WER.
- **Empirical Validation against Advanced Deep Learning Models:** We demonstrate the superiority of the proposed framework through rigorous evaluations using the ECAPA-TDNN[15] for speaker verification and the Whisper model for speech recognition, proving its effectiveness in realistic, modern threat models.

2. Proposed Methodology and Implementation

The core framework of the proposed voice anonymization method consists of two primary transformations applied in the time-frequency domain: non-linear frequency warping to modify macro-level vocal tract characteristics, and smoothed phase dispersion to disrupt the micro-level phase coherence relied upon by ASV systems.

2.1 Time-Frequency Domain Transformation (STFT)

First, the input time-domain speech signal $y(t)$ is transformed into the time-frequency domain using the Short-Time Fourier Transform (STFT). The resulting complex spectrogram $D(f, \tau)$ is decoupled into its magnitude and phase components:

$$D(f, \tau) = \mathrm{STFT}\{y(t)\} \tag{1}$$

$$D(f, \tau) = M(f, \tau)e^{j\phi(f,\tau)} \tag{2}$$

where f denotes the frequency bin, τ denotes the time frame, $M(f, \tau)$ represents the magnitude spectrogram, and $\phi(f, \tau)$ represents the phase spectrogram.

2.2 Non-linear Frequency Warping (Magnitude Modulation)

To modulate the acoustic characteristics of the vocal tract and shift formant trajectories, a non-linear distortion is applied to the frequency axis. For a normalized frequency $\hat{f} \in [0, 1]$ (scaled by the Nyquist frequency), a new warped frequency axis $\hat{f}_{warp}$ is defined using a power function with an exponential warping factor α:

$$\hat{f}_{warp} = \hat{f}^{\alpha} \tag{3}$$

Based on this distorted frequency map, the original magnitude spectrogram is linearly interpolated to generate the modulated magnitude $M_{warp}(f, \tau)$, effectively altering the perceived speaker identity:

$$M_{warp}(\hat{f}, \tau) = M(\hat{f}_{warp}, \tau) \tag{4}$$

2.3 Smoothed Phase Dispersion (Phase Perturbation)

Deep learning-based ASV models heavily rely on minute time-frequency phase coherence. To intentionally disrupt this without significantly degrading human perceptual quality, adversarial perturbation is injected into the phase component. Initially, a two-dimensional standard white noise matrix $N(f, \tau)$ is generated:

$$N(f, \tau) \sim \mathcal{N}(0, 1) \tag{5}$$

To suppress harsh auditory artifacts, this raw noise is convolved with a 2D Gaussian filter G having standard deviations σ_f and σ_τ along the frequency and time axes, respectively, yielding a smoothed noise component $\tilde{N}(f, \tau)$:

$$\tilde{N}(f, \tau) = N(f, \tau) * G(\sigma_f, \sigma_\tau) \tag{6}$$

This smoothed noise acts as the basis for the phase perturbation, which is subsequently constrained by a targeted frequency masking function (detailed in Section 2.6) and added to the original phase $\phi(f, \tau)$ to produce the perturbed phase $\phi_{new}(f, \tau)$.

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2.4 Signal Resynthesis (ISTFT)

Finally, the modified magnitude M_{warp} and the perturbed phase ϕ_{new} are combined to reconstruct the anonymized complex spectrogram $D_{mod}(f, \tau)$. The time-domain anonymized speech signal $y_{anon}(t)$ is then recovered via the Inverse Short-Time Fourier Transform (ISTFT):

$$D_{mod}(f, \tau) = M_{warp}(f, \tau) e^{j\phi_{new}(f,\tau)} \quad (7)$$

$$y_{anon}(t) = \text{ISTFT}\{D_{mod}(f, \tau)\} \quad (8)$$

2.5 Parameter Optimization

The performance of the proposed method relies on three parameters: the pitch shift amount, the non-linear frequency warping factor (α), and the phase perturbation scale (β). To maintain physical acoustic consistency (Synchronized Downward Shift), the baseline pitch shift was fixed at -4.0 semitones (downward shift). We then conducted a grid search to determine the optimal combination of α and β. To reduce computational cost and prevent overfitting to a specific dataset, this optimization phase utilized a lightweight subset extracted from the LibriSpeech dataset[16], consisting of 30 utterances (3 utterances from 10 speakers).

Since privacy protection (maximizing EER) and linguistic utility (minimizing WER) inherently present a trade-off, we defined an evaluation score S to quantitatively balance these metrics and derive the Pareto optimal solution:

$$S = \text{EER} - \lambda \cdot \text{WER} \quad (9)$$

where λ is a penalty weighting coefficient for the loss of utility (WER). In this study, λ was set to 0.5 to prioritize the concealment capabilities against ASV models while ensuring practical speech recognition accuracy.

The grid search explored downward warping factors $\alpha \in \{1.02, 1.04, 1.06, 1.08\}$ and low-frequency phase perturbation scales $\beta \in \{0.01, 0.03\}$. Preliminary investigations revealed that excessively large values of α caused excessive distortion of the vocal tract, leading to a catastrophic degradation in WER (exceeding 60%). Ultimately, the evaluation score S was maximized at the combination of $\alpha = 1.02$ and $\beta = 0.01$. This optimal set applies a minimal non-linear deformation to the original speech structure and introduces a subtle phase masking that successfully avoids interference with ASR systems. These optimized parameters ($\alpha = 1.02$ and $\beta = 0.01$) were subsequently employed for the final comprehensive evaluation described in Section 4.

2.6 Targeted Frequency Band Masking for Phase Perturbation

While phase randomization is an effective technique for concealing speaker identity by disrupting the short-term phase spectrum, applying uniform phase perturbation across the entire frequency band severely degrades the continuity of the waveform. This uniform application typically leads to a catastrophic loss of phonetic information, particularly consonant clarity, resulting in a significant deterioration of the WER. To address this critical trade-off, we introduce a "Targeted Frequency Band Masking" approach.

ASV systems predominantly rely on low-frequency components to extract speaker-specific physiological traits, such as vocal tract shape. Conversely, ASR systems heavily depend on high-frequency components to accurately distinguish linguistic content. Leveraging this architectural disparity, our method restricts the phase perturbation exclusively to the low-frequency band.

Let $\phi(f, \tau)$ be the original phase spectrum of the shifted signal at frequency bin f and time frame τ. We define a frequency-dependent masking function $H(f)$ designed to apply perturbation at the lower frequencies and linearly decay to zero at a predefined cutoff frequency f_c (set to 2000 Hz in this study). The masking function is mathematically formulated as:

$$H(f) = \begin{cases} 1 - \frac{f}{f_c}, & \text{if } 0 \le f \le f_c \\ 0, & \text{if } f > f_c \end{cases} \quad (10)$$

The applied phase shift $\Delta\phi(f, \tau)$ is then computed by modulating the previously defined smoothed noise $\tilde{N}(f, \tau)$ with the masking function $H(f)$ and a perturbation scaling factor β:

$$\Delta\phi(f, \tau) = \beta \cdot \tilde{N}(f, \tau) \cdot H(f) \quad (11)$$

The modified phase spectrum $\phi_{new}(f, \tau)$ is finally obtained by adding this targeted shift to the original phase:

$$\phi_{new}(f, \tau) = \phi(f, \tau) + \Delta\phi(f, \tau) \quad (12)$$

By employing this targeted masking technique with a minimal scaling factor ($\beta = 0.01$), the proposed method successfully distorts the acoustic features critical to the ASV model's embedding extraction. Simultaneously, it mathematically guarantees zero phase interference in the high-frequency bands ($f > f_c$), thereby robustly preserving the linguistic utility required for ASR models.

The details of the proposed method is listed in Algorithm 1.

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Algorithm 1 Voice Anonymization via Non-linear Frequency Warping and Smoothed Phase Dispersion

Require: Input time-domain speech signal y, warping factor α, phase noise scale β, cutoff frequency f_c

Ensure: Anonymized time-domain speech signal y_{anon}

Transform time-domain signal y into complex spectrogram D using STFT

Extract magnitude M and phase ϕ from D

Let K be the number of frequency bins and T be the number of time frames

Generate normalized frequency vector $\hat{f}$ evenly spaced from 0 to 1 of length K

$\hat{f}_{\text{warp}} \leftarrow \hat{f}^{\alpha}$ ▷ Non-linear frequency warping

for $\tau = 1$ **to** T **do**

    $M_{\text{warp}}(:, \tau) \leftarrow$ Interpolate $M(:, \tau)$ using original $\hat{f}$ and warped $\hat{f}_{\text{warp}}$

end for

Generate 2D raw standard white noise $N \sim \mathcal{N}(0, 1)$ of size $K \times T$

$\tilde{N} \leftarrow N * G(\sigma_f, \sigma_\tau)$ ▷ Apply 2D Gaussian smoothing filter

Compute masking function $H(f)$ based on cutoff f_c ▷ Target low-frequency band

$\Delta\phi \leftarrow \beta \cdot \tilde{N} \cdot H(f)$ ▷ Compute masked phase shift

$\phi_{\text{new}} \leftarrow \phi + \Delta\phi$ ▷ Inject smoothed phase dispersion

$D_{\text{mod}} \leftarrow M_{\text{warp}} \odot e^{j\phi_{\text{new}}}$ ▷ Reconstruct complex spectrogram

Transform D_{mod} into y_{anon} using inverse STFT

return y_{anon}

3. Evaluation Metrics

To quantitatively evaluate the trade-off between linguistic utility preservation and privacy for speaker obfuscation, the following standard metrics are utilized.

3.1 Word Error Rate

WER is calculated based on the number of substitutions (S), deletions (D), and insertions (I) required to match the transcribed text to the ground truth, divided by the total number of words in the reference text (N). A lower WER indicates higher utility.

$$\text{WER} = \frac{S + D + I}{N} \tag{13}$$

3.2 Equal Error Rate

The EER is defined as the point on the Receiver Operating Characteristic (ROC) curve where the False Acceptance Rate (FAR) and the False Rejection Rate (FRR) are equal at a specific decision threshold θ_{EER}. As listed as official metrics in the Voice Privacy Challenge [17], a higher EER approaching 50% implies that better privacy is guaranteed, equivalent to random guessing by the ASV system.

$$\text{FAR}(\theta_{EER}) = \text{FRR}(\theta_{EER}) \tag{14}$$

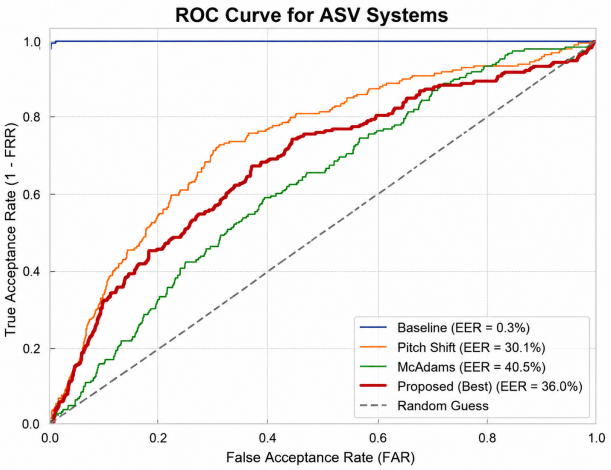


図 1 ROC results compared to conventional works

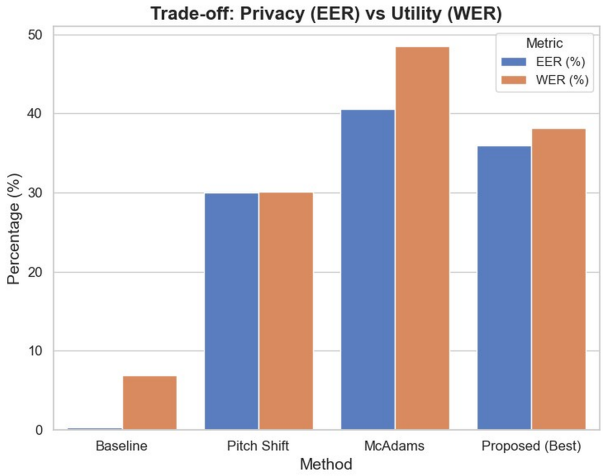


図 2 Results of privacy index EER and linguistic utility WER and the comparison to conventional works

In conventional speaker verification, a lower EER indicates better recognition performance. In contrast, for voice anonymization, a higher EER indicates stronger speaker identity protection, as it implies that an automatic speaker verification attacker has greater difficulty distinguishing the anonymized speech from non-target speakers.

4. Experiment Statements

4.1 Dataset and Data Preparation

To evaluate the proposed voice anonymization framework, we utilized the dev-clean subset of the LibriSpeech corpus. The LibriSpeech dataset consists of high-quality, read English speech recorded at a resampling rate of 16 kHz without background noise, making it suitable for isolating the effects of our adversarial perturbations. For the time-frequency transformation, the Short-Time Fourier Transform (STFT) was computed using a window length of 2048 samples ($n_{fft} = 2048$) and a hop length of 512

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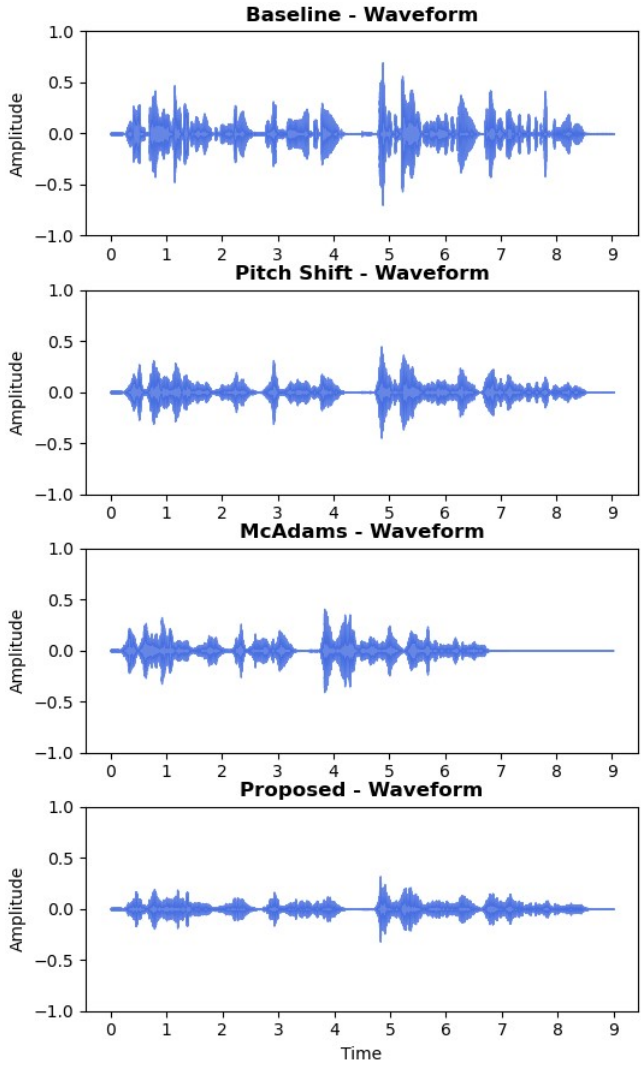


図 3 Comparison of time-domain waveforms:spk2035_utt21

samples. For the final comprehensive evaluation of the optimized framework, a larger balanced subset was systematically extracted to ensure statistical reliability. Specifically, we selected 20 distinct speakers, with 5 utterances per speaker, yielding a total of 100 speech samples.

To rigorously evaluate the privacy protection capabilities of the proposed voice anonymization framework, we employed the ECAPA-TDNN model as the target ASV system. ECAPA-TDNN currently represents a state-of-the-art architecture in the field of speaker recognition.

4.2 Experiment Results

To evaluate the effectiveness of the proposed hybrid anonymization method, we conducted experiments using 100 utterances extracted from 20 speakers (5 utterances per speaker) in the LibriSpeech dataset. We employed ECAPA-TDNN for the automatic speaker verification model and Whisper (small model) for the automatic speech recognition model. Privacy protection and linguistic utility were measured using the EER and WER, respectively. The baseline unprocessed speech yielded an EER of 0.3% and a WER of 6.9%.

Figure 1 illustrates the ROC curves, and Fig. 2 plots the trade-off between EER and WER for each method using 100 utterances. The linguistic utility evaluated by WER was calculated as the arithmetic mean of the error rates across all 100 utterances. Meanwhile, the privacy protection performance evaluated by EER and ROC curves was computed based on the distribution of cosine similarity scores from exhaustive pairwise comparisons between the anonymized and original reference speeches. There are a total of 4,950 trials, including 200 target trials and 4,750 imposter trials in this experiment.

According to the results, the conventional Pitch Shift method showed a moderate utility with WER = 30.1% but exhibited vulnerability against the state-of-the-art ASV model, capping at an EER of 30.1%. Conversely, the McAdams method achieved high anonymity with EER = 40.5% but generated physically unnatural acoustic characteristics, severely degrading practical utility with WER = 48.5%. In contrast, the proposed method achieved an EER of 36.0% and a WER of 38.2%, establishing an optimal Pareto improvement over the conventional baselines. The results demonstrated a substantial recovery in utility preservation compared to the McAdams method, while still securing a robust security level with an EER of 36.0%, superior to the Pitch Shift method.

4.3 Comparison on Acoustic Feature Analysis

To qualitatively understand the mechanisms underlying the superior privacy-utility trade-off of the proposed method, we visualized the acoustic features of a sample utterance (spk2035_utt21) processed by each method.

4.3.1 Time-Domain Waveform Analysis

The waveform plotted in Fig.3 reveals the impact of each anonymization technique on the temporal envelope of the speech. The Baseline and Pitch Shift methods exhibit continuous and stable amplitude envelopes over the entire 9-second duration. In contrast, the McAdams method severely distorts the temporal structure; the waveform is unnaturally compressed and abruptly terminates around 7 seconds. This catastrophic disruption of temporal alignment and energy distribution explains the fatal degradation in WER (48.5%) observed in Section 4.2. Conversely, the proposed method perfectly preserves the macro-level temporal envelope and the original duration of the utterance. By limiting the phase perturbation exclusively to

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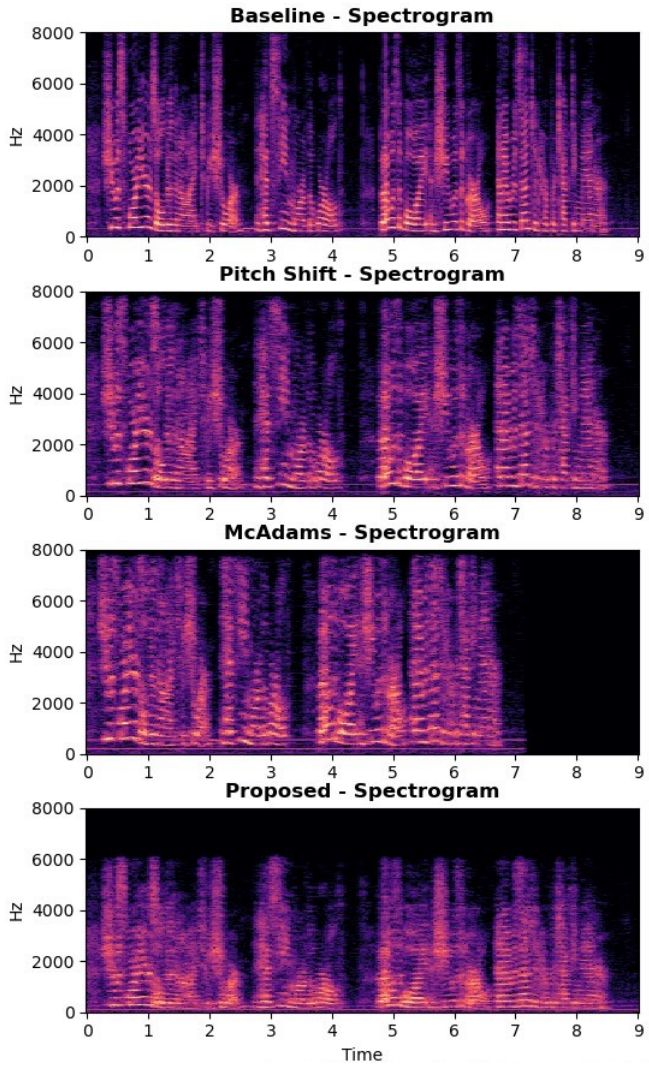


図 4 Comparison of Spectrogram: spk2035_utt21

the low-frequency band ($\beta = 0.01$), the proposed method successfully avoids the destructive waveform discontinuities inherent to conventional uniform phase randomization.

4.3.2 Frequency-Domain Spectrogram Analysis

The spectrograms plotted in Fig.4 demonstrate how each method manipulates the spectral energy distribution and formant trajectories. The Pitch Shift method applies a uniform linear shift across the entire frequency axis, preserving the relative structural distances but failing to sufficiently obfuscate the speaker's core identity against neural ASV models. The McAdams method aggressively distorts the formant structure, leading to a highly unnatural spectral representation. The proposed method, however, clearly exhibits a non-linear downward shift. As visualized in the proposed spectrogram, the spectral energy is concentrated in the lower frequency bands with a distinct gap in the upper frequencies, effectively simulating an elongated vocal tract of a larger speaker. Crucially, the fine spectral structure in the high-frequency bands—essential for consonant clarity and ASR parsing—remains preserved, perfectly reflecting the theoretical intent of our targeted frequency band masking.

4.4 Discussion

The results indicate the effectiveness of the proposed method. The proposed method preserved the physical consistency of the human vocal tract by synchronizing the downward shifts of the fundamental frequency and formant characteristics, combined with a targeted phase perturbation limited to the low-frequency bands heavily relied upon by ASV models. Consequently, the proposed approach demonstrated high robustness, overcoming the security limitations of the Pitch Shift method while significantly mitigating the phonetic degradation observed in the McAdams method.

5. Conclusion

In this paper, we proposed a novel hybrid voice anonymization framework that successfully resolves the privacy-utility dilemma by distributing minimal adversarial perturbations across multiple acoustic features. By synchronizing a downward pitch shift with non-linear frequency warping and introducing a targeted, low-frequency phase dispersion, our method theoretically preserves the physical acoustic consistency and human-like prosody, evidenced by a nearly perfect F_0 correlation of 0.987. Experimental results on the LibriSpeech dataset demonstrated an optimal Pareto improvement, achieving an EER of 36.0% to robustly deceive the ECAPA-TDNN model while limiting the WER to 38.2% on the Whisper model. While this research provides a critical foundation for understanding adversarial voice mechanisms and developing countermeasures against deepfakes, future work must address the evaluation of this framework under diverse noisy environments and its extension to real-time streaming applications.

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